

THE ISOTROPY CLASSIFICATION OF THE TYPE AND TEST MODEL STRUCTURES ON DE MORGAN CUBICAL SETS IN DIMENSIONS UP TO THREE

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ABSTRACT. On presheaves over the De Morgan cube category — the site of CCHM cubical type theory — the weak equivalences of the cubical-type model structure are strictly contained in those of the test model structure: a companion paper produced one explicit separating map, attached to the Klein quotient of the square. This paper turns the single witness into a classification. For every subgroup H of the signed permutation group B_n acting on the n -cube, $n \leq 3$, we determine on which side of the comparison the quotient $\square^n/H$ falls, in terms of two computable invariants of H : its *fixed-cell pattern* and its *realizable character group*. If H is free or has a global fixed cell, the two structures agree on $\square^n/H$. If H is mixed and every realizable character is trivial — twelve subgroups up to $n = 3$, including the full symmetric quotient $\square^3/B_3$ — we construct an explicit map in $W_{\text{test}} \setminus W_{\text{type}}$ attached to $\square^n/H$. The witnesses are produced by a uniform machine, the median collage, in three forms: a two-stratum join, a free resolution by the nerve of H modulo a chosen reflection-free normal subgroup, and horns along maximal strata. The proofs of non-membership in W_{type} are reduced to finite parity certificates; a no-go theorem shows why the resolution is necessary. Beyond the character dichotomy we prove two transfer theorems — separation passes to block products (six further separating subgroups, now with nontrivial characters), and a median deformation retraction onto a twisted diagonal circle forces agreement for every subgroup conjugate into $S_n \times \langle \text{total negation} \rangle$ (eighteen agreements inside the character zone) — and a homological obstruction that decides the last two conjugacy classes. The result is a complete classification in dimensions up to three: for each of the 98 subgroups of B_3 we prove one of two properties — at 78 quotients the two structures assign the same homotopy type, exhibited by explicit structural arguments (free-quotient coverings, contractions, median retractions, and products); at 20 quotients an explicit map attached to the quotient separates the localizers (collage certificates, block-product transfer, and a homological obstruction). All finite combinatorial claims, including the certificates and the mod-2 cohomology computations, are verified by reproducible exhaustive computations.

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1. INTRODUCTION

Let $\square_{\text{DM}}$ be the De Morgan cube category and $\widehat{\square_{\text{DM}}}$ its presheaf category, the site of CCHM cubical type theory [2]. Two homotopy theories live on $\widehat{\square_{\text{DM}}}$: the *cubical-type model structure*, whose fibrant objects are the uniform Kan complexes of type theory, and the *test model structure* of Cisinski [3], with weak equivalences created by the category of elements. The companion papers [6, 7] established the precise relation between their localizers: the containment

$$(U) \quad W_{\text{type}} \subseteq W_{\text{test}}$$

holds [6], and it is strict: [7] exhibits an explicit map $\bar{\Phi} \in W_{\text{test}} \setminus W_{\text{type}}$, built from the De Morgan median interpolation and attached to the quotient $\square^2/K$ of the square by the Klein group of symmetries.

The present paper asks the classification question that the separation makes precise: *for which isotropy quotients does the difference appear?* For a subgroup H of the group B_n of signed coordinate permutations of the n -cube, write $W_H := \square^n/H$ for the strict quotient presheaf. We determine, for every $H \leq B_n$ with $n \leq 3$, whether the two structures agree on W_H — and when they do not, we construct the witness.

The invariants. Two finite invariants of H control the answer. Call an element $h \neq 1$ a *reflection* if it fixes some cell of $\square^n$; by the fixed-cell criterion (Proposition 3.2) this happens exactly when every cycle of the signed permutation h has even sign sum. H is *free* if it has no reflections, *fixed* if some cell is fixed by all of H , and *mixed* otherwise. Second, a character $\chi: H \rightarrow \mathbb{Z}/2$ is *realizable* if some element φ of the free De Morgan algebra $\text{DM}(x_1, \dots, x_n)$ satisfies $\varphi \circ \sigma_h = \neg^{\chi(h)} \varphi$ for all $h \in H$, where σ_h is the substitution action. Realizability is decided by a parity criterion on the middle level of the literal cube (Theorem 3.1), with no search through the (doubly exponential) De Morgan algebra.

What is being classified. Two properties of a quotient must be distinguished, and they are not formal negations of each other.

Definition 1.1. Let X be a presheaf on $\square_{\text{DM}}$.

- (Ag) *The structures agree at X (object level)* if there is a finite zigzag of presheaf-level homotopy equivalences (strict maps with strict cylinder homotopy inverses) connecting X to a *reference object*: a representable, the reversal circle L , or a free quotient $\square^m/V$. At the reference objects the two structures assign the same homotopy type, by contractibility of representables, by (L), and by (FQ) respectively; since presheaf-level homotopy equivalences lie in both localizers, the same then holds at X .
- (Sep) *Separation occurs at X* if some map with codomain X lies in $W_{\text{test}} \setminus W_{\text{type}}$.

Remark 1.2. (Ag) and (Sep) are not complementary by fiat. Property (Sep) transfers along presheaf-level homotopy equivalences (compose the witness with the equivalence), so (Ag) reduces the question “does (Sep) hold at X ?” to the same question at the reference objects — and whether separation can occur at a *contractible* object is equivalent to the open question whether every test-contractible presheaf is type-contractible, which this paper does not address. The classification below is therefore the assignment, to every subgroup, of a property it *provably* enjoys: every H receives (Ag) or (Sep), constructively, and no H is left with neither. For the flagship separating quotients the two properties are in fact known to exclude each other one level deeper: at $\square^2/K$ the object-level homotopy types themselves differ [7], since an abstract equivalence would strictify to a class excluded by the nullity theorem.

Main theorem.

Theorem 1.3 (Classification, $n \leq 3$). *Let $H \leq B_n$, $n \leq 3$.*

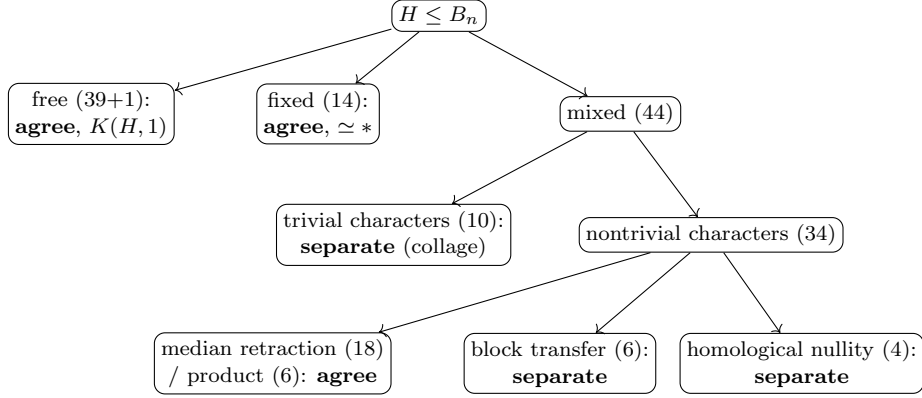
- (1) *If H is free, then W_H is a $K(H, 1)$ in both structures, with the same homotopy type $\mathrm{el}(W_H) \simeq BH$: the structures agree at W_H in the sense of Definition 1.1(Ag), with reference a free quotient.*
- (2) *If H is fixed, then W_H is contractible in both structures: (Ag) with reference a representable.*
- (3) *If H is mixed and every realizable character of H is trivial, then separation occurs at W_H in the sense of Definition 1.1(Sep): there is an explicit map $\bar{\Phi}_H \in \mathbf{W}_{\mathrm{test}} \setminus \mathbf{W}_{\mathrm{type}}$ whose codomain is W_H .*
- (4) *The remaining mixed subgroups (those with a nontrivial realizable character) are also completely classified: at $n = 3$, of the 98 subgroups of B_3 , property (Ag) is proven at exactly 78 quotients and property (Sep) at exactly 20 — the 10 subgroups of case (3), 6 block products of a separating factor with a negation, and 4 subgroups in two further conjugacy classes decided by a homological obstruction (Theorem 11.3).*

Case (3) comprises exactly twelve subgroups in total in dimensions ≤ 3 (actual subgroup counts, not conjugacy classes): the Klein group K and the full group B_2 of the square, and ten subgroups of B_3 of orders 4, 4, 4, 8, 8, 8, 12, 24, 24, 48 — including B_3 itself, so that the type and test structures differ at the full symmetric quotient $\square^3/B_3$.

Cases (1) and (2) hold for all n (Propositions 4.1 and 4.2) and are agreements; the core of the paper is case (3). The remaining class — mixed H with a nontrivial realizable character, of which there are 34 subgroups at $n = 3$ — turns out to split, and the splitting is the second theme of the paper (Section 10): a *retract-transfer theorem* shows that separation passes to block products, giving six separating subgroups with nontrivial characters, so the character criterion does not characterize separation; and a *median-retraction theorem* produces a strict H -equivariant deformation retraction

of $\square^n$ onto a twisted diagonal circle whenever H is conjugate into $S_n \times \langle z \rangle$ (z the total negation, n odd), forcing agreement — eighteen subgroups at $n = 3$, all homotopy equivalent to the reversal circle as presheaves. Together with six further product agreements this settles 94 of the 98 subgroups of B_3 . The final four — two conjugacy classes, resolved by neither transfer — separate by a third device (Section 11): a *homological relative nullity* argument showing that the surviving twisted classes, whose composites factor through aspherical quotients with category of elements $\mathbb{R}\mathbb{P}^\infty$, cannot reach the identity because $\text{el}(W_H)$ has mod-2 cohomology of dimension 2 in a low degree. The classification in dimensions up to three is therefore complete: *agreement at 78 quotients, separation at 20*.

The shape of the classification. The following decision tree summarizes how the status of W_H is determined, with the subgroup counts at $n = 3$; it is the roadmap of the paper.



Total at $n = 3$: agreement at $1 + 39 + 14 + 18 + 6 = 78$ quotients, separation at $10 + 6 + 4 = 20$.

The witness machine. The three forms of the collage, for later reference:

form	model	used for
two-stratum join	$\square^{m_1} \star \square^{m_2} \xrightarrow{\Phi} \square^n$	K, B_2 ; six $H \leq B_3$
E -resolution	$(\square^n \times E_G) \cup \text{horns} \xrightarrow{\Phi} \square^n$	orders 12, 24 (transitive)
poset collage	horns on maximal strata	nested strata; B_3, H_8, H_{24}

All twelve witnesses are produced by one device in three successive generalizations. The *median collage* of [7] interpolates between two parametrized strata of $\square^n$ through the De Morgan median $F(d, a, t) = (d \wedge \neg t) \vee (a \wedge t) \vee (d \wedge a)$; its equivariance reduces to three identities of F (self-duality, swap-reversal, and the face laws). When H has exactly two reflections with disjoint fixed loci, a join of two parameter cubes suffices (the *two-stratum collage*, Theorem 5.4; eight of the twelve witnesses). When the strata form a single H -orbit with larger stabilizers, no finite cube can carry the required action: we prove a no-go theorem (Theorem 7.1) showing that any equivariant model must be cell-free away from reflections, and satisfy this constraint

by tensoring the cube with E_G , the codiscrete nerve of $G = H/V$, where V is a chosen reflection-free normal subgroup of H (the *E-resolution collage*, Theorem 7.3; two more witnesses). For the remaining two subgroups the strata are nested — for B_3 they are six planes containing four lines — and horns along the maximal strata assemble the general *poset collage* (Section 8), whose test-side comparison we show to be independent of the shape of the stratification.

On the type side, non-membership in W_{type} is proven by a *relative nullity theorem*: every map from W_H into a fibrant replacement of the collage is either null-homotopic or factors, already at the strict level, through a quotient $\square^n / \langle \tau \rangle$ by a single reflection — and the latter quotients are contractible in the type structure by the fixed-layer contractions of [6]. The possible factorizations are enumerated by finite *parity certificates* (Section 6): union-find computations over the middle level of the literal cube and over the vertices of the cube, which we run for all twelve subgroups. The certificates are exact — their soundness is the weight-rule construction of Theorem 3.1 — and they replace, in every case, the exhaustive searches through free De Morgan algebras that were still needed in [7].

One step requires input beyond combinatorics: for one of the two order-24 subgroups the fundamental group of $\text{el}(W_H)$ is trivial and non-contractibility must be detected in homology. We show that $\text{el}(W_H)$ is the universal cover of $\text{el}(\square^3/B_3)$ and compute $\dim_{\mathbb{F}_2} H^k(\text{el}(\square^3/B_3); \mathbb{F}_2) = 1, 1, 2$ for $k = 1, 2, 3$ by an explicit cocycle calculation (Section 9); since $\mathbb{RP}^\infty$ has $\dim H^3 = 1$, the cover is not contractible. As a byproduct the calculation shows that $\text{el}(\square^3/B_3)$ is a 2-local space: we prove that the relevant coefficient module — the first homology of the incidence graph of the diagonal lines and planes of the cube — is cohomologically trivial over the 3-Sylow subgroup.

Relation to the literature. The comparison problem originates in the folklore argument of Sattler and Coquand (2018) that cubical models with reversal do not present spaces [5], revisited in the introduction of Awodey–Cavallo–Coquand–Riehl–Sattler [1]; [6] completed the key computation of that argument and showed that no object-level witness among the quotients of the square exists, and [7] produced the map-level witness. The classification perspective of the present paper has a precedent in the equivariant program of [1] for cartesian cubes, where symmetric quotients obstruct the classical model structure and are repaired equivariantly; on the De Morgan site the phenomenon is finer — the obstruction is invisible at the level of objects (both structures see the same weak-contractibility verdicts on all isotropy quotients of the square [6]) and the classification must separate localizers through maps. The invariants that decide it — fixed-cell patterns and realizable characters — appear to be new.

Outline. Section 2 fixes notation and imports. Section 3 develops the parity criteria and the classification tables. Section 4 proves the agreement cases.

Section 5 constructs the two-stratum collage and proves Theorem 5.4; Section 6 introduces the certificates. Section 7 proves the no-go theorem and the E -resolution theorem. Section 8 treats nested strata and B_3 . Section 9 contains the cohomology computation. Section 12 records the machine verification; Section 13 discusses the open zone.

2. PRELIMINARIES

2.1. The site and the two structures. $\square_{\text{DM}}$ has objects $[n]$ and morphisms generated by faces, degeneracies, and the De Morgan operations $\wedge, \vee, \neg, 0, 1$; $\square_{\text{DM}}([m], [1]) = \text{DM}(x_1, \dots, x_m)$ is the free De Morgan algebra. A cell of a presheaf X at level $[m]$ is an element of $X([m])$; cells of $\square^n$ at level $[m]$ are n -tuples over $\text{DM}(x_1, \dots, x_m)$. The signed permutation group $B_n = (\mathbb{Z}/2)^n \rtimes S_n$ acts on $\square^n$: $h = (\pi, s)$ acts on cells by $(h \cdot c)_i = \neg^{s_i} c_{\pi(i)}$ (the *cell action*) and on $\text{DM}(x_1, \dots, x_n)$ by substitution σ_h . For $H \leq B_n$ the quotient $W_H = \square^n/H$ is the levelwise orbit presheaf.

We use the test model structure and the cubical-type model structure on $\widehat{\square_{\text{DM}}}$ *only* through the following eight imported statements; every later proof cites them by these labels, and no other result of the companion papers is used.

- (U) The containment $W_{\text{type}} \subseteq W_{\text{test}}$ (U) [6].
- (W2) The computation of $\text{el}(\square^2/K)$ by the stabilizer stratification, including the fibered arrow-category argument for the comma pieces [6].
- (L) $L = \mathbb{I}/\neg$ is a $K(\mathbb{Z}/2, 1)$ in both structures [6].
- (FQ) The free-quotient theorem: for cell-free H , the algebraic small object argument commutes with the quotient and $\square^n/H$ is a $K(H, 1)$ in both structures [6].
- (FL) The fixed-layer contractions: $\square^2/\langle \text{sw} \rangle$ and its twisted variant are contractible in the type structure, by explicit connection homotopies at the presheaf level [6].
- (Collage) The sieve-cosieve collage formula: for a functor preserving a sieve decomposition, the nerve of the category of elements is the homotopy colimit of the sieve, cosieve, and comma parts, naturally in the functor [7].
- (Strict) The strictification lemma: $\text{Hom}(W_H, Y) \cong \{s \in Y([n]) : s \cdot h = s \ \forall h\}$ for fibrant Y [7].
- (R) The formal fibrant replacement $R(-)$ by Garner's algebraic small object argument for uniform fibrations, with its stage filtration, the descent lemma for quotient boxes, and the equivariant filler-slide lemma [7].

Fixed versions of both companion manuscripts are supplied with this submission, so the referee can verify each import against a stable text.

2.2. Fixed cells, strata, characters. A cell c at any level is *fixed* by h if $h \cdot c = c$. The *fixed locus* $\ell_h \subseteq \square^n$ of a reflection h is the subpresheaf of

cells fixed by h ; it is abstractly a subcube $\square^m$ with m the number of cycles of h , parametrized by signed monomials (propagate signs along each cycle from its minimal slot). *Strata* of H are the fixed loci of its reflections; H permutes them by conjugation. Call a normal subgroup $V \trianglelefteq H$ *reflection-free* if no nontrivial element of V is a reflection. For such a V the quotient $X_V := \square^n/V$ is a $K(V, 1)$ in both structures by the free-quotient theorem, so that $W_H = X_V/(H/V)$ re-bases the problem over an aspherical object. We emphasize that reflection-free normal subgroups are *not* closed under join, so there is in general no largest one: already in B_2 , the negation group $\langle n_x, n_y \rangle \cong (\mathbb{Z}/2)^2$ and the rotation group $\langle r \rangle \cong \mathbb{Z}/4$, $r(x, y) = (\neg y, x)$, are both normal and reflection-free (every nontrivial element has an odd-signed cycle), yet together they generate all of B_2 , which contains the reflection sw . Accordingly, whenever a construction below uses a reflection-free normal subgroup, its statement carries the choice of V as data; for each group treated in this paper we exhibit the choice explicitly.

3. THE PARITY CRITERIA

The *literal cube* $L_n = \{0, 1\}^{2n}$ has points $p = (v_{x_1}, v_{\neg x_1}, \dots)$; an element of $\text{DM}(x_1, \dots, x_n)$ is exactly a monotone map $L_n \rightarrow \{0, 1\}$ (the up-set representation, [6]). Substitution σ_h acts on points by permuting literal pairs with swaps for signs; De Morgan negation acts by $(\neg\varphi)(p) = 1 - \varphi(\rho p)$ where ρ swaps each literal pair and complements. Both actions preserve or reflect the *weight* $w(p)$ (number of ones): σ_h^* preserves it and ρ sends w to $2n - w$. The *middle level* is the antichain $\{w = n\}$.

Theorem 3.1 (Exactness). *Let $\chi: H \rightarrow \mathbb{Z}/2$ be a character. Consider the parity constraint graph on the middle level of L_n : for each $h \in H$ and each middle point q , an edge from q to $\sigma_h^* q$ of parity $\chi(h)$, composed with ρ when $\chi(h) = 1$. Then χ is realizable if and only if the graph is parity-consistent (no odd cycle).*

Proof. Necessity: a realizing φ restricted to the middle level satisfies the edge relations, and an odd cycle forces $\varphi(q) = 1 - \varphi(q)$. Sufficiency is the *weight rule*: set $\varphi = 1$ on $w > n$, $\varphi = 0$ on $w < n$, and choose any parity-consistent assignment on the middle level. Off the middle level the constraints hold automatically — an edge with $\chi(h) = 1$ composes with ρ and hence flips the weight across n , matching the 1/0 prescription, while an edge with $\chi(h) = 0$ preserves the weight. Monotonicity is free because the middle level is an antichain lying between the all-zero lower part and the all-one upper part. $\square$

Proposition 3.2 (Fixed-cell criterion). *$h \neq 1$ fixes some cell if and only if every cycle of h has even sign sum. More generally a subgroup H has a global fixed cell iff the slot union-find with sign parities induced by all $h \in H$ is consistent.*

Proof. A fixed cell satisfies $c_i = \neg^{s_i} c_{\pi(i)}$; propagating around a cycle gives $c = \neg^{\Sigma s} c$, impossible for odd sums since free De Morgan algebras have no self-dual elements (a point p with $pp = p$, e.g. $(1, 0)$ in each pair, forces $\varphi(p) = 1 - \varphi(p)$). For even sums, assign an arbitrary element to one slot per cycle and propagate. The subgroup statement is the simultaneous version. $\square$

The same middle-level method decides, for any homomorphism ν from H to a group of signed permutations of a finite coordinate set, whether a ν -equivariant tuple of De Morgan elements exists: this *vector parity criterion* runs union-find over (coordinate, middle point) pairs and is again exact by the componentwise weight rule. It is the engine of the certificates in Section 6.

Proposition 3.3 (Tables). *Up to $n \leq 3$ the criteria give: for $n = 2$, ten subgroups — one trivial, five free, two fixed, two mixed (K and B_2), both with trivial realizable characters. For $n = 3$, 98 subgroups — one trivial, 39 free, 14 fixed, 44 mixed, of which exactly ten have only trivial realizable characters. All counts are machine-verified twice: once by the criteria and once, for $n = 2$, by exhaustive enumeration of $\text{DM}(x, y)$.*

4. THE AGREEMENT FAMILIES

Proposition 4.1 (Free case). *If H is free then W_H is a $K(H, 1)$ in the cubical-type structure and $\text{el}(W_H) \simeq BH$; the structures agree on W_H .*

Proof. This is the free-quotient theorem of [6]: the algebraic small object argument commutes with quotients by groups acting freely on cells (box shapes are cell-connected, so unique box lifting descends), making $\square^n \rightarrow W_H$ a principal covering with contractible total object on both sides. $\square$

Proposition 4.2 (Fixed case). *If H has a global fixed cell then W_H is contractible in both structures.*

Proof. If H has a global fixed cell it has a global fixed vertex $v \in \{0, 1\}^n$: the slot union-find of Proposition 3.2 is consistent, so set a representative coordinate of each class to 0 and propagate the signs; the resulting 0/1-tuple is fixed by every $h \in H$. Now contract $\square^n$ to v coordinatewise:

$$c_t(x)_i = \begin{cases} x_i \wedge \neg t & \text{if } v_i = 0, \\ x_i \vee t & \text{if } v_i = 1. \end{cases}$$

This is H -equivariant precisely because v is a fixed vertex: for $h = (\pi, s)$ with $s_i = 1$ one has $v_i = 1 - v_{\pi(i)}$, and De Morgan duality $\neg(x \wedge \neg t) = \neg x \vee t$ converts the $v_{\pi(i)}$ -branch into the v_i -branch, while for $s_i = 0$ the branches agree. The contraction is a strict cylinder homotopy from the identity to the constant v , descends to W_H , and exhibits W_H as type-contractible; test contractibility follows by (U). $\square$

5. THE TWO-STRATUM COLLAGE

Throughout this section $H \leq B_n$ is mixed with exactly two reflections $h_1 \neq h_2$, whose loci $\ell_i = p_i(\square^{m_i})$ are disjoint (equivalently $\langle h_1, h_2 \rangle$ has no global fixed cell, Proposition 3.2); disjointness forces $h_i^2 = 1$. All realizable characters of H are assumed trivial.

Construction 5.1 (Two-stratum join collage). Let $J_H := \square^{m_1} \star \square^{m_2}$, the double mapping cylinder with cells $(\vec{d}, \vec{a}, t)$ and the two end collapses at $t = 0, 1$, and define $\Phi: J_H \rightarrow \square^n$ componentwise by

$$\Phi_j = F(p_{1,j}(\vec{d}), p_{2,j}(\vec{a}), t).$$

H acts on the coordinates of J_H by the *twisted action* ν : a stratum-preserving h acts on each parameter block by the signed permutation induced on the locus parameters, fixing t ; a stratum-swapping h interchanges the blocks and reverses t .

Lemma 5.2. Φ is H -equivariant: $h \circ \Phi = \Phi \circ \nu(h)$. Consequently Φ descends to $\bar{\Phi}_H: J_H/\tilde{H} \rightarrow W_H$.

Proof. For sign twists this is the self-duality $\neg F(x, y, t) = F(\neg x, \neg y, t)$; for the stratum swap it is the reversal identity $F(x, y, \neg t) = F(y, x, t)$, immediate from the symmetric median form $(x \wedge \neg t) \vee (y \wedge t) \vee (x \wedge y)$; the end collapses are respected by the face laws $F(x, y, 0) = x$, $F(x, y, 1) = y$. All three are identities of De Morgan algebras, verified additionally by machine. $\square$

Lemma 5.3 (Formula-level structure of the collapse loci). *Let $Z_i \subseteq J_H$ be the subpresheaf of cells whose Φ -image is h_i -fixed. Then (i) Z_i is closed under the flow $t \mapsto t \wedge \neg s$ (respectively $t \vee s$), by the flow identity $F(x, y, t \wedge \neg s) = F(x, F(x, y, t), \neg s)$; (ii) $Z_1 \cap Z_2 = \emptyset$, since a common cell would map into $\ell_1 \cap \ell_2$, the fixed subpresheaf of $\langle h_1, h_2 \rangle$, empty by hypothesis; (iii) the deck group acts freely on interior cells (no self-dual elements).*

Theorem 5.4 (Two-stratum separation). *Under the standing hypotheses, suppose moreover that*

- (1) *the certificate of Section 6 holds for H , and*
- (2) *$\text{el}(W_H)$ is not weakly contractible.*

Then $\bar{\Phi}_H \in W_{\text{test}} \setminus W_{\text{type}}$. Both hypotheses hold for all instances with $n \leq 3$: (1) by machine verification, and (2) by the two criteria in the last part of the proof below.

We prove the two sides separately, the test side through three lemmas. Throughout, write $\varphi = \text{el}(\bar{\Phi}_H)$, let $S_W \subseteq W_H$ be the union of the (images of the) two strata — a subpresheaf, hence a sieve in $\text{el}(W_H)$ — and let $S'' = \varphi^{-1}(S_W)$ be the pulled-back sieve in $\text{el}(J_H/\tilde{H})$; by (Collage) both nerves are homotopy colimits of the sieve, cosieve, and comma parts, and φ

induces a map of colimit diagrams:

$$(1) \quad \begin{array}{ccccc} NS'' & \longleftarrow & N\widehat{W}'' & \longrightarrow & NU'' \\ \varphi|_{S''} \downarrow & & \downarrow \varphi| & & \downarrow \varphi|_{U''} \\ NS_W & \longleftarrow & N\widehat{W} & \longrightarrow & NU_W \end{array}$$

where $\widehat{W}$ denotes the comma part. The three lemmas below show that the three vertical maps — the *legs* — are weak equivalences; since both rows present the nerves as homotopy colimits, this proves the test side of Theorem 5.4. The same diagrammatic template is reused for the E -resolution collage (Section 7), the poset collage (Section 8), and the cohomology computations (Section 9). Write $H^+ \leq H$ for the stratum-preserving subgroup (index ≤ 2).

Lemma 5.5 (Leg A: sieve parts). *φ restricts to a weak equivalence $S'' \rightarrow S_W$.*

Proof. A cell of $J_H/\tilde{H}$ lies in S'' iff its Φ -image lies in a stratum, i.e. iff its representative lies in $Z_1 \cup Z_2$ (Lemma 5.3); the two parts are disjoint (Lemma 5.3(ii)) and exchanged or preserved according to the coset of H^+ , so $S'' = (Z_1 \sqcup Z_2)/\tilde{H} \cong Z_1/\tilde{H}^+$ when the strata form one orbit, and splits into two halves otherwise; the same description holds for S_W with Z_i replaced by ℓ_i . The flow $t \mapsto t \wedge \neg s$ preserves Z_1 (Lemma 5.3(i)), is natural in the cell variables, and is compatible with the quotient relations because the twisted action fixes t on H^+ ; it therefore defines a presheaf cylinder homotopy contracting $Z_1/\tilde{H}^+$ onto the strict end $t = 0$, on which $\tilde{\Phi}_H$ restricts to the isomorphism $[\vec{d}] \mapsto [p_1(\vec{d})]$ of end-quotients (the residual actions match by construction of ν). Hence each component of S'' deformation retracts onto a component of S_W mapped isomorphically, and $\varphi|_{S''}$ is a weak equivalence. $\square$

Lemma 5.6 (Leg C: cosieve parts). *φ restricts to a weak equivalence between the complements $U'' \rightarrow U_W$ of the sieves.*

Proof. U_W is the full subcategory of $\text{el}(W_H)$ on the cells not in the strata. Its cover — the corresponding full subcategory of $\text{el}(\square^n)$ — has terminal object $\text{id}_{[n]}$ (the generic cell is in no stratum), hence is contractible, and H acts freely on it: a nontrivial stabilizer of a cell is generated by reflections fixing it, and cells fixed by h_i lie in ℓ_i . Thus $NU_W \simeq BH$ as the quotient of a contractible free H -category. On the collage side, U'' is covered by the full subcategory of non-collapse cylinder cells of J_H , again with terminal object (the top cylinder cell, which is not in $Z_1 \cup Z_2$ since the strata are proper), and the deck group acts freely on cylinder cells by Lemma 5.3(iii); so $NU'' \simeq BH$ as well. The upstairs comparison Φ is equivariant along the identity of H and maps one contractible free cover to the other, so the induced map of quotients is a $BH \rightarrow BH$ equivalence (it is a map of principal H -quotients of contractible categories inducing the identity on H). $\square$

Lemma 5.7 (Leg B: comma parts). *φ restricts to a weak equivalence of the comma parts of the two collages.*

Proof. The comma part of the collage for $\text{el}(W_H)$ is the category of pairs (stratum cell s , non-stratum cell u , a map $s \rightarrow u$ in $\text{el}(W_H)$); by the fibered arrow-category argument of (W2) it is equivalent to $\text{Arr}(\ell\text{-cover}, U\text{-cover})/H$, where the arrow category fibers over the contractible stratum cover with contractible fibers (each has a terminal object given by the identity cell) and H acts freely through the U -component. Hence the comma part is $B(\text{Stab})$ for the stabilizer of a stratum component — BH^+ when the strata form one orbit, one copy per stratum otherwise. The same computation applies verbatim on the collage side, with the stratum cover replaced by Z_i -covers (contractible by the flow of Lemma 5.5) and the U -cover by the cylinder cells; and φ maps the one arrow category to the other equivariantly over the identification of the stabilizers, hence induces an equivalence of quotients as in Lemma 5.6. $\square$

Proof of Theorem 5.4. Test side. By (Collage), $N\text{el}$ of both sides is the homotopy colimit of the three parts, and Lemmas 5.5–5.7 provide a levelwise weak equivalence of the diagrams; hence $\text{el}(\bar{\Phi}_H)$ is a weak equivalence and $\bar{\Phi}_H \in \mathcal{W}_{\text{test}}$.

Type side. Suppose $R(\bar{\Phi}_H)$ were a homotopy equivalence. Choosing a homotopy inverse and strictifying by (Strict), we obtain an H -invariant class $g \in R(J_H/\tilde{H})([n])$ with $R(\bar{\Phi}_H) \circ g \simeq \text{unit}_{W_H}$. We show every invariant class is null-homotopic; then the unit is null, $\text{id}_{R(W_H)}$ is null, and W_H is type-contractible, contradicting hypothesis (2) via (U). We induct over the stage filtration of R from (R).

Stage 0, interior classes. An invariant class represented by a cylinder cell c satisfies $c \circ \sigma_h = \nu(\delta(h)) \cdot c$ for a unique $\delta(h)$ in the deck group (uniqueness by Lemma 5.3(iii)), and δ is a homomorphism. By hypothesis (1) — the certificate — no nontrivial δ admits a solution, so $\delta = 1$ and c is strictly invariant; then c descends to a map $W_H \rightarrow J_H \rightarrow J_H/\tilde{H}$, and J_H is type-contractible (a join of representables, contracted by the connection homotopies), so the class is null.

Stage 0, end classes. A class landing in the end $\bar{\ell}_i$ is a cell of the stratum quotient, i.e. a tuple of De Morgan elements satisfying $u \circ \sigma_h = \gamma(h) \cdot u$ for the residual twist γ valued in the pure-sign residual group of the stratum. Each sign component of γ is then a character of H realized by a component of u (Theorem 3.1); by triviality of realizable characters, $\gamma = 1$, the cell is a strict tuple, and it lifts along the representable cover $\square^{m_i} \rightarrow \bar{\ell}_i$; maps through a representable are null by the connection contraction.

Stage 0, horn and degenerate classes. The end-adapted slides $t \mapsto t \wedge \neg z$ and $t \mapsto t \vee z$ (matched across the swap coset by $\neg(t \wedge \neg z) = \neg t \vee z$) are compatible with the quotient relations by the equivariant filler-slide lemma of (R), and they push any horn or cylinder-boundary class to an end class,

reducing to the previous case. Degenerate classes factor through lower levels and are null by the same cases one level down.

Higher stages. Let s be an invariant class represented by a fresh filler value of stage $\beta + 1$. We do *not* argue that the invariance of the value forces strict invariance of its box datum — in the uniform algebraic replacement, fresh filler values are identified by the strict naturality relations, so an invariant value can arise from a non-invariant box (this is precisely the subtlety addressed in [7]). Instead we apply the imported equivariant filler-slide lemma (R) directly: the slide is a cylinder homotopy, compatible with both generating relations of the quotient presentation and with the twisted group action (the end-adapted form matches the two slide directions across the swap coset, as above), from s to a class s_1 supported on the open-box boundary, i.e. on material of stage $\leq \beta$; and the slide preserves invariance. The stage induction hypothesis then applies to s_1 , and $s \simeq s_1$ is null.

Non-contractibility (hypothesis (2)) for $n \leq 3$. If the two strata lie in one H -orbit, $\pi_1 \text{el}(W_H) \twoheadrightarrow H / \langle \langle h_1, h_2 \rangle \rangle \neq 1$ (for the instances at hand this quotient has order 2: the reflections have even total sign, so their normal closure lies in the kernel of the sign-sum character). If the strata are separately invariant, $H = \langle h_1, h_2 \rangle$ is a Klein four-group and $\text{el}(W_H) \simeq \text{hocolim}(B\mathbb{Z}/2 \leftarrow B(\mathbb{Z}/2)^2 \rightarrow B\mathbb{Z}/2)$ has $H_3 = \mathbb{Z}/2 \neq 0$ exactly as for the Klein quotient of the square (W2). $\square$

At $n \leq 3$ the hypotheses of Theorem 5.4 hold for K and B_2 (recovering [7] and its dihedral extension) and for six subgroups of B_3 (orders 4, 4, 4, 8, 8, 8), giving the first eight witnesses of Theorem 1.3(3).

6. THE PARITY CERTIFICATES

Definition 6.1. Let a model J with twisted H -action ν on a finite coordinate set be given. For a homomorphism $\delta: H \rightarrow H$, the δ -system is the vector parity system for $c \circ \sigma_h = \nu(\delta(h)) \cdot c$ over the middle level of L_n (Section 3), together with, when the model carries an E_G -factor (Section 7), the *vertex system*: a G -valued union-find over the vertices of $\square^n$ for $\varepsilon \circ (h \text{ on vertices}) = \chi(\delta(h)) \cdot \varepsilon$. The *certificate* for (H, J) asserts: every δ with nontrivial coordinate action has an inconsistent δ -system.

The certificate is a finite, exact computation (soundness by the weight rule; the vertex system is exact since E_G is 0-coskeletal, so its cells are vertex labellings). Its meaning: *no twisted interior class exists*, so all interior classes are strict and null. Two structural remarks. First, the certificate is implied by character triviality whenever the deck action is componentwise (no coordinate permutation): each sign character of δ is then realized by a component, and the swap character by the t -coordinate. This proves the certificate for the Klein case abstractly. Second, the converse correlation is exact in the two-stratum family at $n \leq 3$: among the fifteen two-stratum

subgroups of B_3 , the certificate holds precisely for the six with trivial realizable characters (machine verification) — for mixed subgroups with nontrivial characters, unobstructed twists exist, and Sections 10 and 11 determine what they mean: either genuine equivalences (the median retractions) or classes obstructed homologically.

7. THE E -RESOLUTION COLLAGE

When the reflections of H form larger orbits the two-stratum model is unavailable, and product models fail for a reason worth recording.

Theorem 7.1 (No-go). *Let Z be any presheaf with an H -action admitting an equivariant map to $\square^n$. Then every non-reflection $v \in H$ acts cell-freely on Z . In particular the product $\prod_i \square^{m_i}$ of the stratum parameter cubes, with the permutation action induced by an H -orbit of strata, admits no equivariant map to $\square^n$ whenever some non-reflection acts on the strata with an even-signed cycle — which happens for every involution, since $v^2 = 1$ forces even cycle signs.*

Proof. A v -fixed cell of Z would map to a v -fixed cell of $\square^n$, and non-reflections have none (Proposition 3.2). For the product, the diagonal cells along an even-signed cycle are fixed (Proposition 3.2 again, applied to the parameter action). $\square$

The resolution: free the action on an auxiliary factor. Let $V \trianglelefteq H$ be a chosen reflection-free normal subgroup, $G = H/V$, $\chi: H \twoheadrightarrow G$, and let E_G be the codiscrete nerve of G : the 0-coskeletal presheaf with $E_G([m]) = G^{\text{vertices}([m])}$, contractible, levelwise finite, with G acting freely. Every reflection has $\chi(h) \neq 1$, because $\chi(h) = 1$ means $h \in V$ and V is reflection-free by definition; so reflections act freely on E_G , while non-reflections act freely on $\square^n$: the product $\square^n \times E_G$ is cell-free throughout.

Construction 7.2 (E -resolution collage). Suppose V acts transitively on the strata of H , with stratum representatives $\ell_i = p_i(\square^m)$. Set

$$J_H := (\square^n \times E_G) \cup \bigcup_i (\ell_i \times E_G \times \square^1),$$

the body with one horn per stratum, glued to the body along $t = 0$ via $\ell_i \times E_G \subseteq \square^n \times E_G$ and collapsing at $t = 1$ to the bare stratum ℓ_i (forgetting E_G). Φ is the projection to $\square^n$ on the body and the constant interpolation on horns. Diagrammatically, J_H is the pushout

$$\begin{array}{ccc} \prod_i \ell_i \times E_G & \hookrightarrow & \square^n \times E_G \\ \downarrow t=0 & & \downarrow \\ \prod_i \ell_i \times E_G \times \square^1 & \longrightarrow & J_H \xrightarrow{\Phi} \square^n \end{array}$$

followed by the end collapse $(\cdot, e, 1) \sim (\cdot)$ identifying the outer end of the i -th cylinder with the bare ℓ_i : the E_G -factor lives on the body, where it

enforces freeness, and is forgotten exactly at the strata, where the isotropy is meant to survive.

Theorem 7.3 (Single-stratum separation). *Let H be mixed with trivial realizable characters, let $V \trianglelefteq H$ be a reflection-free normal subgroup acting transitively on the strata, put $G = H/V$, and suppose the certificate holds for (H, J_H) and $\text{el}(W_H)$ is not weakly contractible. Then $\bar{\Phi}_H \in \mathcal{W}_{\text{test}} \setminus \mathcal{W}_{\text{type}}$. The hypotheses hold for the two remaining transitive subgroups of B_3 , with explicit choices of V : the order-12 group ($\cong A_4$, four diagonal lines, $V = V_{\text{even}}$ the even-sign negation Klein group, $G = \mathbb{Z}/3$) and an order-24 group ($V = V_{\text{even}}$ again, G of order 6); in both cases V is normal, reflection-free, and transitive on the four lines by direct verification.*

Proof. Test side. We verify the three legs of the collage comparison as in Theorem 5.4. *Leg (A).* The sieve S'' on the collage side consists of the horns together with the body cells over the strata ($\ell_i \times E_G$ -cells: their Φ -images lie in the strata, and no other body cell maps there since Φ is the projection). Each horn is a cylinder collapsing at $t = 1$ to the bare stratum, so the cylinder direction is a presheaf deformation retraction of S'' onto the union of the strict ends; the retraction is well-defined across the quotient because the twisted action preserves t and permutes horns. The ends assemble on the nose to the strata part of W_H (the end of the i -th horn is ℓ_i , and the end identifications are exactly the H -identifications), and $\bar{\Phi}_H$ restricts there to the identity. *Leg (C).* The cosieve cover on the collage side is (free cells of $\square^n \times E_G$: contractible (a product of a category with terminal object and the contractible E_G) and H -cell-free (non-reflections act freely on the first factor, reflections on the second, since $\chi(h) \neq 1$ for reflections). On the quotient side the cover is the free-cell category, contractible with free H -action as before. Both quotients are BH and the projection Φ induces an equivalence as in Lemma 5.6. *Leg (B).* The comma fibers over the disjoint union of the stratum covers $\ell_i \times E_G$ (contractible each), with contractible fibers; H permutes the components transitively (by transitivity of V on strata) with component stabilizer $\text{Stab}(\ell_1)$, acting freely through the cosieve component. Hence the comma is $B(\text{Stab}(\ell_1))$ on both sides, compatibly.

Type side. Strictify by (Strict) and induct over stages as before. *Interior (body) classes:* a twisted class is a pair (u, ε) with $u \circ \sigma_h = \delta(h) \cdot u$ on the $\square^n$ -factor and $\varepsilon \circ \sigma_h = \chi(\delta(h)) \cdot \varepsilon$ on the E_G -factor; the certificate (the vector parity system for u together with the G -valued vertex system for ε — exact because E_G is 0-coskeletal, so its cells are vertex labellings) excludes all nontrivial δ ; for the order-12 group all 32 nontrivial homomorphisms among the 33 endomorphisms are obstructed, for the order-24 group all 57 of 58. Strict classes descend along the body $\square^n \times E_G$, which is type-contractible, hence are null. We record the crucial role of the E_G -factor: the bare body $\square^n$ would admit the tautological section $W_H \rightarrow \square^n/H$, but an equivariant $\varepsilon: \square^n \rightarrow E_G$ would have to assign to a g -fixed vertex (which exists for every

reflection g) a $\chi(g)$ -twisted-fixed value in G , contradicting freeness of the G -action; so no section survives into the resolved body. *End and horn classes:* exactly as in Theorem 5.4 — residual twists on the single end split into sign characters of H , excluded by character triviality; strict end cells lift to the representable cover of the stratum; horn classes slide to the end. *Higher stages:* by the equivariant filler-slide lemma (R), exactly as in Theorem 5.4: an invariant fresh filler value slides invariantly to earlier-stage material and the stage induction applies.

Non-contractibility. For the order-12 group, the collage decomposition gives $\text{el}(W_H) \simeq \text{hocolim}(* \leftarrow B\mathbb{Z}/3 \rightarrow BH)$ (the stratum quotient is a bare interval, contributing a point; the comma is $B(\text{Stab}) = B\mathbb{Z}/3$), and Mayer–Vietoris yields $H_1 = 0$ (the Sylow-3 inclusion is an isomorphism on first homology, $H_1(A_4) = \mathbb{Z}/3$) and $H_2(\text{el}W_H) \cong H_2(BA_4) = \mathbb{Z}/2 \neq 0$. For the order-24 group, $\pi_1(\text{el}W_H) = H/\langle\langle\text{reflections}\rangle\rangle = \mathbb{Z}/2$ by the even-sign argument. $\square$

8. NESTED STRATA, THE POSET COLLAGE, AND B_3

The last two subgroups — an order-24 group and B_3 itself — have nested strata: six diagonal planes containing four diagonal lines (each line in three planes; twelve incidences). Two candidate models fail: iterated joins carry no strict symmetric action (associators are homotopies, not identities), and product centers fail by Theorem 7.1. The correct model attaches horns only along *maximal* strata:

$$J_H := (\square^n \times E_G) \cup \bigcup_{P \text{ maximal}} (P \times E_G \times \square^1),$$

and the key point is that the test-side comparison is insensitive to the shape of the stratification:

Proposition 8.1. *For the poset collage, $\text{el}(\bar{\Phi}_H)$ is a weak equivalence for every mixed H , irrespective of the shape of the stratification poset.*

Proof. Leg (A). The pulled-back sieve consists of the horn cells together with the body cells over the strata. Define the retraction by the horn cylinder direction: a horn cell (c, e, t) over a maximal stratum P retracts to its end value $c \in P$, and a body cell over the strata retracts through the horn it borders. The only point requiring care is a body cell c over a *shared lower stratum* $D \subset P \cap P'$: it borders the horns of both P and P' , and the two retractions send (c, e) to the same end value c , because the collapse $(c, e, 1) \sim c$ forgets e and is defined by the globally defined projection, not by a choice of horn. Hence the retraction is well-defined and natural, and the sieve deformation retracts onto the union of the strict ends. That union is exactly the strata part $S_W \subseteq W_H$: the end of the P -horn is P itself, and the identifications among ends (within and across strata, including along shared lower strata) are precisely the H -identifications defining S_W . On the ends $\bar{\Phi}_H$ restricts to the identity of S_W .

Leg (C). The cosieve cover on the collage side is (cells over no stratum) $\times E_G$, contractible with cell-free H -action (non-reflections free on the first factor, reflections on the second since χ kills no reflection); on the quotient side it is the free-cell category with its free H -action. Both quotients are BH and Φ induces the identity comparison as in Lemma 5.6.

Leg (B). The comma category fibers over the cover of the sieve part with contractible fibers. Here the base is no longer a disjoint union of contractible pieces — the strata covers intersect along lower strata — but the argument of Lemma 5.7 needs only that the two sides have the *same* base: on both sides the base is the category of elements of the strata union upstairs (times the contractible E_G on the collage side, which changes nothing), and H acts freely through the cosieve component. Both comma parts are therefore the Borel construction of the same H -category, and Φ induces the identity comparison. Three equivalent legs give the claim by (Collage). $\square$

On the type side the certificate for B_3 *fails*: of the 400 endomorphisms of B_3 , six nontrivial twists are unobstructed, and they are precisely $\delta = \iota_\tau \circ \chi$ for a character $\chi: B_3 \rightarrow \mathbb{Z}/2$ and a plane reflection τ (three plain coordinate swaps and three swaps composed with double negation). This failure is not a defect but the correct structure:

Theorem 8.2 (Relative nullity). *For the poset collage of B_3 , every strict class $W \rightarrow R(J/\tilde{H})$ is null-homotopic or carries one of the six twists above; and for a twisted class g , the composite $R(\bar{\Phi}) \circ g$ factors, strictly, as*

$$W \longrightarrow \square^3/\langle\tau\rangle \longrightarrow W \longrightarrow R(W),$$

through the quotient by the single reflection τ .

Proof. Enumeration of twists. Body classes of the poset collage are pairs (u, ε) as in Theorem 7.3, with twist $\delta \in \text{Hom}(B_3, B_3)$. Enumerating all 400 endomorphisms and running both parity systems leaves, besides the trivial homomorphism, exactly the six twists $\delta = \iota_\tau \circ \chi$ named above (machine verification; the images are the two-element subgroups generated by the six plane reflections).

Strict classes. For $\delta = 1$ the class descends along the type-contractible body $\square^3 \times E_G$ and is null, exactly as before.

Twisted classes factor through $\square^3/\langle\tau\rangle$. Let g carry the twist $\iota_\tau \circ \chi$. Its body component $u: \square^3 \rightarrow \square^3$ satisfies $u \circ \sigma_h = \tau^{\chi(h)} \circ u$, i.e. u is equivariant along χ for the $\langle\tau\rangle$ -action on the target; hence u descends to $\bar{u}: W \rightarrow \square^3/\langle\tau\rangle$, and by construction $\bar{\Phi} \circ g$, computed on the body, is the composite of $\bar{u}$ with the projection $\square^3/\langle\tau\rangle \rightarrow W$ followed by the unit of R . This is the claimed strict factorization. (The ε -component contributes only within the contractible E_G -direction and does not affect the class of the composite.)

End classes. A class landing in a plane end $\bar{P}$ is a plane cell twisted by the residual group of P , which is the pure-sign Klein group — there is no parameter swap, by the stabilizer computation: the setwise stabilizer has order 8, the pointwise stabilizer order 2, and all four residual actions are

componentwise signs. The twist therefore splits into two independent character systems in the sense of Theorem 3.1, and triviality of the realizable characters of B_3 forces both to vanish; the strict cell lifts along the representable cover $\square^2 \rightarrow \bar{P}$ and is null. The line end is handled the same way with a single character (the reversal residual). Horn classes slide to end classes; higher stages reduce to earlier stages by the equivariant filler-slide lemma (R), as in Theorem 5.4. $\square$

Theorem 8.3 (Separation at the full symmetric quotient). $\bar{\Phi}_{B_3} \in W_{\text{test}} \setminus W_{\text{type}}$; the type and test structures differ at $\square^3/B_3$.

Proof. Test membership is Proposition 8.1. If $R(\bar{\Phi})$ had a homotopy inverse, strictification would produce a class g with $R(\bar{\Phi}) \circ g \simeq$ the unit. Null classes are excluded because the unit of a contractible replacement would make W type-contractible, contradicting $\pi_1 \text{el}(W) = B_3 / \langle \langle \text{reflections} \rangle \rangle = \mathbb{Z}/2$ (all reflections have even total sign, so the normal closure lies in the kernel of the negation-sum character; machine computation confirms equality) together with (U). Twisted classes are excluded by Theorem 8.2: their composites factor through $\square^3 / \langle \tau \rangle \cong (\square^2 / \langle \text{sw} \rangle) \times \square^1$ (or its twisted variant), which is type-contractible by the fixed-layer contractions of [6], so the composite is null, and the unit is not. $\square$

For the order-24 nested subgroup H' the certificate holds outright (all nontrivial twists obstructed), but $\pi_1 \text{el}(W_{H'}) = 1$ and non-contractibility needs homology; this is the subject of the next section.

9. A COHOMOLOGY COMPUTATION

Lemma 9.1. H' contains all fourteen reflections of B_3 , hence every cell stabilizer; the residual $\mathbb{Z}/2$ acts freely on the cells of $W_{H'}$ and $\text{el}(W_{H'})$ is the universal cover of $\text{el}(\square^3/B_3)$.

Thus $\text{el}(W_{H'})$ is contractible if and only if $\text{el}(\square^3/B_3) \simeq \mathbb{RP}^\infty$, which we refute by computing mod-2 cohomology through the two-level collage decomposition

$$\begin{array}{ccc} & C & \\ \swarrow & & \searrow \\ \text{el}(S) & & BB_3 \end{array} \quad \begin{array}{ccc} & BG_e & \\ \swarrow & & \searrow \\ BG_\ell & & BG_P \end{array}$$

where S is the strata part, the left triangle presents $\text{el}(\square^3/B_3)$ as its homotopy colimit, and the right triangle presents the comma C (Proposition 9.3). The Mayer–Vietoris map whose ranks decide everything is

$$\Phi_d: H^d(\text{el}S) \oplus H^d(BB_3) \longrightarrow H^d(C),$$

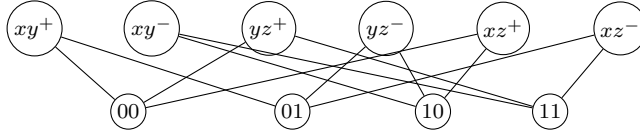
the difference of the two restrictions.

Proposition 9.2. $\text{el}(S) \simeq \mathbb{RP}^\infty \times \mathbb{RP}^\infty$.

Proof sketch. The strata part has its own collage over lines and planes: the line piece is $B\mathbb{Z}/2$ (reversal residual), the plane piece is $B(\mathbb{Z}/2)^2$ (pure-sign residual acting freely; the pointwise stabilizer acts trivially and disappears in the strict quotient), and the comma piece is $B\mathbb{Z}/2$ generated by the total negation z , mapping isomorphically to the line residual and diagonally into the plane residual. A double mapping cylinder with one leg an equivalence collapses onto the other end. $\square$

Proposition 9.3. $C \simeq \text{hocolim}(BG_\ell \leftarrow BG_e \rightarrow BG_P)$ with $|G_\ell| = 12$, $|G_e| = 4$, $|G_P| = 8$ the stabilizers of a line, an incidence, and a plane; equivalently the Borel construction on the incidence graph T_0 of lines and planes, a wedge of three circles.

Proof. The comma fibers over the cover of the strata union with contractible fibers and free B_3 -action through the cosieve component, as in Lemma 5.7; the cover of the strata union is, up to the contractions of the individual strata, the incidence poset of the four diagonal lines and six diagonal planes. Its nerve T_0 is the incidence graph drawn below: bipartite on $4 + 6$ vertices with 12 edges (each line lies in three planes, each plane contains two lines; the three “parallel” plane pairs are the only non-incidences), connected, with first Betti number $12 - 10 + 1 = 3$. Lines are labelled by the sign pattern (e_2, e_3) of $(a, \neg^{e_2}a, \neg^{e_3}a)$; planes by the identified pair and its sign.



The Borel construction decomposes over the orbit poset — one orbit each of lines, planes, and incidences, with stabilizers of orders 12, 8, 4 — giving the double mapping cylinder description. $\square$

Proposition 9.4. $M := H_1(T_0) \cong \mathbb{Z}^3$ is cohomologically trivial over the 3-Sylow subgroup $\langle g \rangle$ of B_3 . Consequently $C \rightarrow BB_3$ is a 3-local homology equivalence and $\text{el}(\square^3/B_3)$ has no odd-primary reduced homology.

Proof. Let g be the even-signed 3-cycle of coordinates. On T_0 , g fixes exactly one vertex — the main-diagonal line — and no plane vertex: it permutes the other three lines cyclically and the six planes in two 3-cycles, and the twelve edges fall into four free orbits. Choosing the spanning tree given by breadth-first search from the fixed line and taking the three fundamental cycles of the remaining chords as a basis of $M \cong \mathbb{Z}^3$, the action of g is the integral matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ -1 & -1 & 0 \\ 1 & 1 & 1 \end{pmatrix}, \quad A^3 = I, \quad \text{tr}(A) = \text{tr}(A^2) = 0,$$

so the character of M is that of the regular representation of $\mathbb{Z}/3$. Cohomological triviality is now a direct computation with A . The norm $N =$

$I + A + A^2$ has rows $(0, 0, 0)$, $(0, 0, 0)$, $(1, 2, 3)$, so $NM = \mathbb{Z}e_3 = M^g$ (the fixed lattice is spanned by e_3 , and $x + 2y + 3z$ attains 1): the Tate group $\hat{H}^0 = M^g/NM$ vanishes. For $H_1(\mathbb{Z}/3; M) = \ker N/\text{im}(A - I)$: both lattices are the rank-2 sublattice $\{x + 2y + 3z = 0\}$ — the two columns $(-1, -1, 1)$ and $(1, -2, 1)$ of $A - I$ satisfy the equation and have index one in it (their coordinates in the basis $(1, 1, -1), (2, -1, 0)$ of the kernel form a matrix of determinant -1) — so $H_1 = 0$, and by the periodicity of cyclic group cohomology all higher groups vanish. Since the reduced homology of the fiber T_0 of the Borel fibration $C \rightarrow BB_3$ is concentrated in degree one and equals M , and $H_p(B_3; M)_{(3)}$ is computed by the invariants of the 3-Sylow with vanishing higher terms, the map $C \rightarrow BB_3$ is a homology isomorphism after localization at every odd prime (only $p = 3$ divides $|B_3|$ oddly). The strata side of the collage is built of 2-groups' classifying spaces, hence is 3-locally trivial; the Mayer–Vietoris sequence then shows $\text{el}(\square^3/B_3)$ has trivial reduced homology with $\mathbb{Z}_{(p)}$ -coefficients for all odd p . $\square$

Theorem 9.5. $\dim_{\mathbb{F}_2} H^k(\text{el}(\square^3/B_3); \mathbb{F}_2) = 1, 1, 2$ for $k = 1, 2, 3$. In particular $\text{el}(\square^3/B_3) \not\cong \mathbb{RP}^\infty$, $\text{el}(W_{H'})$ is not contractible, and the separation pipeline applies to H' .

Proof. Conventions. All cohomology is with $\mathbb{F}_2$ -coefficients, computed on the full (not normalized) inhomogeneous bar complex: a d -cochain on a finite group G is a function $G^d \rightarrow \mathbb{F}_2$, with the usual simplicial coboundary; cup products of characters are $(\alpha \smile \beta)(g, h) = \alpha(g)\beta(h)$ and $(\alpha \smile f)(g, h, k) = \alpha(g)f(h, k)$. Cocycle and coboundary spaces are computed by exact Gaussian elimination over $\mathbb{F}_2$ (the implementation is validated on $(\mathbb{Z}/2)^2$ and $\mathbb{Z}/4$, reproducing $\dim H^2 = 3$ and 1). Restriction along a homomorphism is precomposition of cochains. The groups are realized concretely as signed permutation groups: $G_\ell \cong S_3 \times \langle z \rangle$ (order 12), $G_e = \langle \tau, z \rangle \cong (\mathbb{Z}/2)^2$, $G_P = \langle \tau, n_1, n_2 \rangle \cong (\mathbb{Z}/2)^3$ (a plane's setwise stabilizer with pointwise part τ and pure-sign residual generators n_1, n_2), and B_3 itself. The full rank tables are collected in Appendix A.

The Mayer–Vietoris frame. The collage gives $\text{el}(\square^3/B_3)$ as the double mapping cylinder of $\text{el}(S) \leftarrow C \rightarrow BB_3$, so $\dim H^d(\text{el}(\square^3/B_3)) = \dim \ker \Phi_d + \dim \text{coker } \Phi_{d-1}$, where $\Phi_d: H^d(\text{el}S) \oplus H^d(BB_3) \rightarrow H^d(C)$ is the difference of restrictions, and by Proposition 9.3 the target decomposes through $H^d(BG_\ell) \oplus H^d(BG_P)$ compatibly over $H^d(BG_e)$.

Degrees 1 and 2. Dimensions: H^1 of the pieces are 2, 2 ($\text{el}S, BB_3$) mapping to $H^1(C)$ of dimension 3; the map has full rank 3, giving $\dim H^1 = 1$ (the negation-sum character, matching $\pi_1 = \mathbb{Z}/2$) and zero cokernel. In degree 2 the dimensions are 3 and 4 over a target of dimension 6 ($\dim H^2$ of G_P, G_ℓ, B_3 being 6, 3, 4), and elimination of the explicit cocycles shows Φ_2 has full rank 6; hence $\dim H^2 = (3 + 4) - 6 = 1$.

Degree 3. The target $H^3(C)$ has dimension 10. The images of the strata side (4 cubic classes) and of the decomposable classes of BB_3 (products of the two characters with the four H^2 -classes) span a subspace of dimension

exactly 9. The only potentially missing contribution is the indecomposable part of $H^3(BB_3)$, which is pulled back from the rotation subgroup S_4 ; rather than identify it, we pull back the *entire* 3-cocycle space $Z^3(S_4)$ — of dimension $554 = 551 + 3$, matching $\dim B^3 = 551$ and $\dim H^3(S_4; \mathbb{F}_2) = 3$ — along both composites $G_\ell \rightarrow S_4$ and $G_P \rightarrow S_4$ (each element maps to its rotation part), and reduce all 554 basis vectors against the 9-dimensional span and the coboundaries: none enlarges the span. Hence $\text{rank } \Phi_3 = 9$ and $\dim H^3 = (4 + 7) - 9 = 2 \neq 1 = \dim H^3(\mathbb{RP}^\infty; \mathbb{F}_2)$, where 4 and 7 are the dimensions of $H^3(\text{el}S)$ and of $H^3(BB_3; \mathbb{F}_2)$. $\square$

This completes the proof of Theorem 1.3.

10. BEYOND THE CHARACTER CRITERION

The thirty-four mixed subgroups of B_3 with a nontrivial realizable character are not uniform: both outcomes occur among them, by the two theorems of this section.

Theorem 10.1 (Retract transfer). *Let the coordinates split into H -invariant blocks with $H = H' \times H''$ acting blockwise, so that $W_H = W_{H'} \times W_{H''}$. If the structures differ at $W_{H'}$, they differ at W_H : if Φ' is a witness for H' , then $\Phi' \times \text{id}$ is one for H .*

Proof. $W_{H''}$ has a global point v (a vertex orbit) and a terminal map $W_{H''} \rightarrow \square^0$, giving $i = \text{id} \times v$ and $p = \text{id} \times !$ with $p \circ i = \text{id}$. Writing J' for the domain of Φ' , the same maps exhibit Φ' as a retract of $\Phi' \times \text{id}_{W_{H''}}$ in the arrow category:

$$\begin{array}{ccccc} J' & \xrightarrow{i} & J' \times W_{H''} & \xrightarrow{p} & J' \\ \Phi' \downarrow & & \downarrow \Phi' \times \text{id} & & \downarrow \Phi' \\ W_{H'} & \xrightarrow{i} & W_{H'} \times W_{H''} & \xrightarrow{p} & W_{H'} \end{array}$$

with horizontal composites the identities. The middle arrow lies in $\mathbf{W}_{\text{test}}$ (its category of elements is a product of a weak equivalence with an identity up to the standard product comparison). If it lay in $\mathbf{W}_{\text{type}}$, then so would its retract Φ' , since the weak equivalences of a model structure are closed under retracts — contradicting the hypothesis. $\square$

Applied to the block products $K \times \langle \neg z \rangle$ and $B_2 \times \langle \neg z \rangle$ and their conjugates, Theorem 10.1 yields six separating subgroups of B_3 of orders 8 and 16 *with nontrivial realizable characters*: the character criterion is sufficient but not necessary for separation.

Theorem 10.2 (Median retraction). *Let n be odd and $H \leq S_n \times \langle z \rangle$ (z the total negation) be mixed. Let $M = M(x_1, \dots, x_n)$ be the n -ary median (the symmetric, self-dual majority polynomial). Then the flow*

$$c_t = (F(x_1, M, t), \dots, F(x_n, M, t))$$

is a strict H -equivariant cylinder homotopy from the identity to the retraction onto the diagonal, and it exhibits a presheaf-level homotopy equivalence $W_H \simeq L$. In particular the two structures agree at W_H , both assigning the homotopy type $B\mathbb{Z}/2$. The same holds for every H conjugate in B_n to such a subgroup, by transport along the conjugating automorphism of $\square^n$.

Proof. Equivariance for a permutation part is the symmetry of M ; for the uniform sign it is self-duality of M and of F in its first two arguments. The face laws give $c_0 = \text{id}$ and $c_1 = (M, \dots, M)$, the retraction onto the diagonal; idempotence $M(u, \dots, u) = u$ makes the diagonal a section of the character map $\bar{\varphi} = M: W_H \rightarrow L$ (the diagonal descends to L because H acts on it through $\langle z \rangle$, and it is all of the image since H mixed forces an element with z -part). A strict cylinder homotopy is a homotopy in both model structures, so $\bar{\varphi}$ is an equivalence in both. All identities are verified by machine. $\square$

Two remarks. First, the failure of Theorem 10.2 at even n is precisely the non-existence of a symmetric self-dual median of even arity — which is the parity lemma of [7] that drives the Klein separation: the two phenomena are the same fact seen from both sides. Second, at $n = 3$ a conjugacy analysis shows that eighteen subgroups fall under Theorem 10.2 (six literally, twelve after conjugation), six agree as block products (fixed factor times L), and six separate by Theorem 10.1; of the thirty-four, exactly four subgroups — two conjugacy classes, of orders 8 and 24 — are resolved by neither method. They are the subject of the next section.

11. THE LAST TWO CLASSES: HOMOLOGICAL RELATIVE NULLITY

The two remaining classes are $H_8 = V_{\text{even}} \rtimes \langle \text{sw} \rangle$ (the even-sign negation Klein group extended by a swap; a dihedral group of order 8, three conjugates) and $H_{24} = (\mathbb{Z}/2)^3 \rtimes \mathbb{Z}/3$ (all negations extended by a rotation). Both sit exactly on the frontier: H_8 contains the Klein group K and H_{24} contains the A_4 -type group, both separating, but not as block factors; neither is conjugate into $S_3 \times \langle z \rangle$. H_8 satisfies the two-stratum hypotheses of Section 5 except character triviality (two disjoint plane strata swapped by the odd-sign coset); H_{24} carries the E -resolution model of Section 7 (with $V = (\mathbb{Z}/2)^3$, reflection-free, normal, and transitive on the four lines). In both cases the certificate leaves survivors, all with image of order two: for H_8 , two reflection images (sw and its twisted variant) and three free images (the negation pairs); for H_{24} , seven free images (the seven nontrivial negations).

Theorem 11.1 (Homological relative nullity). *Let g be a class twisted by δ with image $F = \langle \nu \rangle$ of order two. If ν is a reflection, the composite $R(\bar{\Phi}) \circ g$ factors through the type-contractible $\square^3/\langle \nu \rangle$ and is null. If ν is free, the composite factors through $\square^3/F$, a $K(\mathbb{Z}/2, 1)$ with $\text{el} \simeq \mathbb{RP}^\infty$; if it were homotopic to the unit, then applying el (legitimate on type-homotopies and, via*

the containment (U), on type-equivalences) would exhibit $H^*(\text{el}(W_H); \mathbb{F}_2)$ as a split submodule of $H^*(\mathbb{RP}^\infty; \mathbb{F}_2)$, forcing $\dim H^k \leq 1$ for all k . The same bound follows if the identity factors through an end piece, whose category of elements is again $\mathbb{RP}^\infty$ (the residual quotients of the strata are $L \times \square^1$ and L).

Theorem 11.2. *For $k = 1, 2, 3$:*

$$\dim_{\mathbb{F}_2} H^k(\text{el}(W_{H_8}); \mathbb{F}_2) = 1, 1, 2, \quad \dim_{\mathbb{F}_2} H^k(\text{el}(W_{H_{24}}); \mathbb{F}_2) = 1, 2, 4.$$

Consequently every surviving class is obstructed, and separation occurs at W_{H_8} and $W_{H_{24}}$.

Proof. Collage decompositions. For H_8 the two plane strata form one orbit with setwise stabilizer K and pointwise stabilizer $\langle \text{sw} \rangle$; the residual $K/\langle \text{sw} \rangle$ acts freely on the plane parameters, so the stratum piece is $\text{el}(L \times \square^1) \simeq B(K/\langle \text{sw} \rangle)$, the comma is BK (arrow-category argument with component stabilizer K), and the free part is $BH_8 = BD_4$:

$$\text{el}(W_{H_8}) \simeq \text{hocolim}(B(K/\langle \text{sw} \rangle) \leftarrow BK \rightarrow BD_4).$$

For H_{24} the four lines form one orbit with stabilizer $\mathbb{Z}/6 = \langle \text{rot} \rangle \times \langle z \rangle$, pointwise $\mathbb{Z}/3$, residual the reversal:

$$\text{el}(W_{H_{24}}) \simeq \text{hocolim}(B\mathbb{Z}/2 \leftarrow B\mathbb{Z}/6 \rightarrow BH_{24}).$$

Ranks. With the conventions of Theorem 9.5 the Mayer–Vietoris ranks are computed on explicit cocycles; the dimension tables and ranks are in Appendix A. For H_8 : piece dimensions $(1, 2, 2)$, $(1, 3, 3)$, $(1, 4, 4)$ in degrees 1, 2, 3 (stratum, comma, free part), ranks 2, 3, 3, giving $\dim H^k = 1, 1, 2$. For H_{24} : piece dimensions $(1, 1, 1)$, $(1, 1, 2)$, $(1, 1, 4)$, ranks 1, 1, 1, giving $\dim H^k = 1, 2, 4$; here $\dim H^k(BH_{24}; \mathbb{F}_2) = 1, 2, 4$ is the dimension of $\mathbb{Z}/3$ -invariants of $\mathbb{F}_2[a, b, c]$ in degree k (the Sylow-2 subgroup $(\mathbb{Z}/2)^3$ is normal, so stable elements are invariants; the invariant counts are the numbers of rotation-orbits of monomials).

Conclusion. Fundamental groups are $\mathbb{Z}/2$ in both cases (the normal closures of the reflections have index two, by the even-sign argument), giving non-contractibility. The bound of Theorem 11.1 fails at degree 3 for H_8 and already at degree 2 for H_{24} ; every surviving twist has image of order two (reflection images for two of the five H_8 -survivors — killed by the contractible-quotient case — free images for the rest), and end classes factor through stratum pieces whose categories of elements are again $\mathbb{RP}^\infty$; so no class can compose to the unit, and the separation argument concludes as in Theorem 5.4. $\square$

This closes the classification:

Theorem 11.3 (Complete classification at $n = 3$). *Of the 98 subgroups of B_3 , property (Ag) of Definition 1.1 holds at 78 quotients — the trivial group, 39 free, 14 fixed, 6 block products of a fixed factor with L , and 18 median-retractable subgroups — and property (Sep) holds at the remaining*

20: the 10 mixed subgroups with trivial realizable characters, 6 block-product transfers, and the 4 subgroups of the classes H_8 and H_{24} .

12. MACHINE VERIFICATION

All finite claims are verified by exhaustive computation (not by a proof assistant; the mathematical proofs are in the text): the median identities and equivariance; the classification tables of Proposition 3.3 for $n = 2$ (twice: criteria and enumeration of $\text{DM}(x, y)$) and $n = 3$ (criteria); the census of interior invariant classes for K and B_2 (168^3 states, zero essential classes, cross-checking the certificates); the δ -certificates for all separating subgroups (including all 33, 58, 58, 400, 36, 72 endomorphisms of the groups of orders 12, 24, 24, 48, 8, 24); the stabilizer and stratum data; the incidence graph and the module computations of Proposition 9.4; and the cohomology ranks of Theorem 9.5, Theorem 11.2, and Appendix A.

The scripts, together with the fixed versions of the companion manuscripts [6, 7], are archived with the artifact [8]: release `v1.15.0`, commit `1fe3a15`, version DOI 10.5281/zenodo.21899704. The principal scripts, each a self-contained Python program requiring no dependencies beyond NumPy, are: `parity_index.py` and `parity_index3.py` (classification tables, criteria (C1)–(C3)); `strata3.py` (stratum data); `collage_type.py` and `collage_type_g2.py` (median identities, equivariance, censuses); `delta_obstruction.py` (two-stratum certificates); `single_stratum_cert.py` (E -resolution certificates); `certificates3.py` (the certificates with survivor images for all six groups of Appendix A, with the choices of V verified normal and reflection-free); and `coho_lib.py` with the driver `coho_ranks.py` (cocycle spaces, all three Mayer–Vietoris rank tables, the $Z^3(S_4)$ pullback). Each script runs by `python3 <script>` and prints the quantities quoted in the text; total runtime is minutes on commodity hardware.

13. OUTLOOK

Two observations organize the completed table. First, in every agreement case the agreement is established by an explicit presheaf-level construction — a free-quotient covering, a contraction, a median retraction, or a product of these: agreement, where it holds, holds for elementary reasons. Second, in every separation case the witness or obstruction is detected by finite parity data together with at most a low-degree mod-2 cohomology computation of $\text{el}(W_H)$: no infinite or transfinite invariant was ever needed. This asymmetry — soft agreements, finitely certified separations — is the main structural lesson of the classification, and it suggests that the dichotomy is decidable in all dimensions.

Conjecture 13.1. *For every n : (i) mixed $H \leq B_n$ with only trivial realizable characters separate, with collage witnesses; (ii) separation is inherited by block products; (iii) mixed subgroups conjugate into $S_n \times \langle z \rangle$ (n odd)*

agree by the median retraction; (iv) for every mixed H the status of W_H is decided by the certificate together with the homological relative nullity of Theorem 11.1, so that the classification is effectively computable in every dimension.

Parts (ii) and (iii) are proven here for all n (odd n in (iii)). Part (i) is reduced to the general certificate (established in the componentwise case and verified for all instances with $n \leq 3$) and to non-contractibility of $\text{el}(W_H)$, for which the covering and cohomology methods of Sections 9 and 11 appear to generalize. Beyond the classification itself, the finer homotopy types — such as $\text{el}(\square^3/B_3)$ with its mod-2 cohomology $1, 1, 2$ — and the type-side homotopy types of the separating quotients remain to be determined; the latter is the natural continuation of the program.

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Data availability. The verification scripts supporting the finite claims are openly available in the artifact repository [8].

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Use of AI tools. An AI assistant was used in the preparation of this manuscript; the author takes full responsibility for its content.

APPENDIX A. COMPUTATIONAL DATA FOR THE COHOMOLOGY ARGUMENTS

All computations are over $\mathbb{F}_2$ on full bar cochains (conventions in the proof of Theorem 9.5). For each Mayer–Vietoris frame $\text{el}(S) \leftarrow C \rightarrow U$ we list the dimensions of H^d of the three pieces and the rank of the difference map Φ_d ; the resulting $\dim H^d(\text{el}W) = \dim \ker \Phi_d + \dim \text{coker } \Phi_{d-1}$ is in the last column. (In every case below Φ_{d-1} is surjective, so the cokernel term vanishes.)

$W = \square^3/B_3$. Pieces: $\text{el}(S) \simeq \mathbb{RP}^\infty \times \mathbb{RP}^\infty$; C via $H^d(BG_\ell) \oplus H^d(BG_P)$ over $H^d(BG_e)$ with $|G_\ell| = 12$, $|G_e| = 4$, $|G_P| = 8$; $U = BB_3$.

d	$H^d(\text{el}S)$	$H^d(C)$	$H^d(U)$	$\text{rank } \Phi_d$	$H^d(\text{el}W)$
1	2	3	2	3	1
2	3	6	4	6	1
3	4	10	7	9	2

Supporting dimensions: $\dim H^2$ of G_P, G_ℓ, B_3 are 6, 3, 4; $\dim H^3$ are 10, 4, 7; $\dim H^d(BG_e) = d + 1$. The degree-3 rank uses the pullback of the full cocycle space $Z^3(S_4)$ of the rotation subgroup, of dimension $554 = 551 + 3$ ($\dim B^3(S_4) = 551$, $\dim H^3(S_4; \mathbb{F}_2) = 3$); all 554 basis vectors reduce into the 9-dimensional span of the decomposable and strata-side images.

$W = W_{H_8}$. Frame: $B(K/\langle \mathbf{sw} \rangle) \leftarrow BK \rightarrow BD_4$.

d	$H^d(B\mathbb{Z}/2)$	$H^d(BK)$	$H^d(BD_4)$	$\text{rank } \Phi_d$	$H^d(\text{el}W)$
1	1	2	2	2	1
2	1	3	3	3	1
3	1	4	4	3	2

$W = W_{H_{24}}$. Frame: $B\mathbb{Z}/2 \leftarrow B\mathbb{Z}/6 \rightarrow BH_{24}$.

d	$H^d(B\mathbb{Z}/2)$	$H^d(B\mathbb{Z}/6)$	$H^d(BH_{24})$	$\text{rank } \Phi_d$	$H^d(\text{el}W)$
1	1	1	1	1	1
2	1	1	2	1	2
3	1	1	4	1	4

(Entries are $\mathbb{F}_2$ -dimensions in all three tables.)

Certificate statistics. For each separating group, the number of endomorphisms $|\text{Hom}(H, H)|$, the number of nontrivial twists surviving both parity systems, and their images:

H	$ \text{Hom} $	surv.	images
K, B_2 , six two-stratum $H \leq B_3$	16–36	0	—
order-12 (A_4)	33	0	—
order-24 (four lines)	58	0	—
order-24 (nested)	58	0	—
B_3	400	6	$\langle \tau \rangle$, τ a plane reflection
H_8	36	5	$\langle \mathbf{sw} \rangle, \langle \mathbf{sw} \cdot \mathbf{n} \rangle$; three free $\mathbb{Z}/2$
H_{24}	72	7	seven free $\mathbb{Z}/2$

REFERENCES

- [1] S. Awodey, E. Cavallo, T. Coquand, E. Riehl, C. Sattler. The equivariant model structure on cartesian cubical sets. *Advances in Mathematics* 495 (2026), 110965.
- [2] C. Cohen, T. Coquand, S. Huber, A. Mörtberg. Cubical type theory: a constructive interpretation of the univalence axiom. *21st International Conference on Types for Proofs and Programs (TYPES 2015)*, LIPIcs 69, 5:1–5:34, 2018.
- [3] D.-C. Cisinski. *Les préfaisceaux comme modèles des types d’homotopie*. Astérisque 308, Soc. Math. France, 2006.
- [4] G. Maltsiniotis. La catégorie cubique avec connexions est une catégorie test stricte. *Homology, Homotopy and Applications* 11 (2009), 309–326.
- [5] C. Sattler. Do cubical models of type theory also model homotopy types? Lecture, Hausdorff Institute workshop “Types, Homotopy Type theory, and Verification”, Bonn, 2018.
- [6] R. Shibusawa. The test comparison for De Morgan cubical sets: reversal, equivariance, and the join of two projective spaces. Submitted, 2026. Fixed version archived with the artifact [8].
- [7] R. Shibusawa. An explicit separation of the type and test model structures on De Morgan cubical sets. Submitted, 2026. Fixed version archived with the artifact [8].
- [8] R. Shibusawa. Artifact repository `cubical-strict-j`. <https://github.com/r-shibusawa/cubical-strict-j>, release v1.15.0, commit 1fe3a15, archived at the version DOI 10.5281/zenodo.21899704 (concept DOI for all versions: 10.5281/zenodo.21405961); the release also archives the fixed versions of [6, 7] and all scripts of Section 12.

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